

masDouble Integration method :-

$$EI \frac{d^2 y}{dx^2} = m = f(x)$$

$$EI \frac{dy}{dx} = \int f(x) + C_1$$

$$EI y = \int f(x) + C_1 x + C_2$$

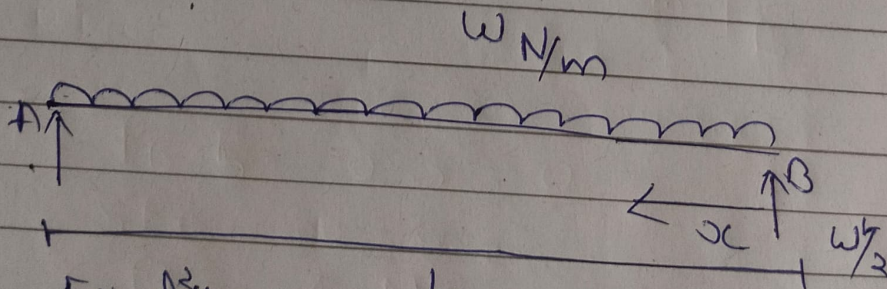
H.W~~Cantilever with udl~~
cantilever

cantilever with udl

" " Point

Simply supported with udl

" " Point



$$EI \frac{d^2 y}{dx^2} = m = \frac{wLx}{2} - \frac{wx^2}{2}$$

$$EI \frac{dy}{dx} = \frac{wLx^2}{4} - \frac{wx^3}{6} + C_1$$

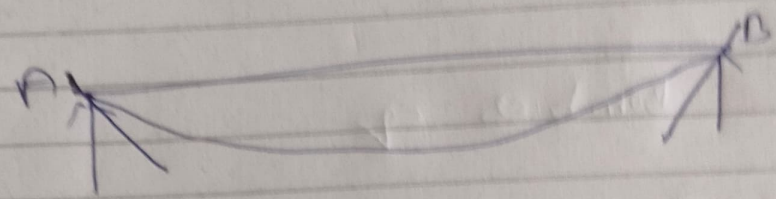
$$EI y = \frac{wLx^3}{12} - \frac{wx^4}{24} + C_1 x + C_2$$

$$x=0, \quad y=0$$

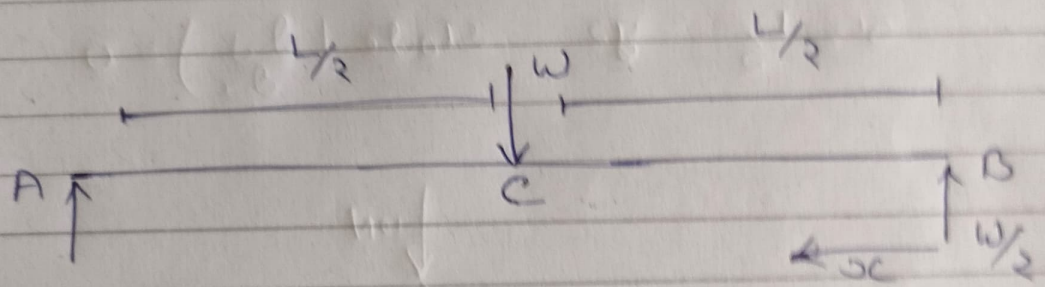
$$x=L, \quad y=0$$

$$x=\frac{L}{2} \quad \text{or} \quad y \text{ max}$$

step -



simply supported with point load :-



$$EI \frac{d^2 y}{dx^2} = M = -\frac{W \times L}{2} \times \frac{L}{2} + \frac{W \times L}{2}$$

$$EI \frac{d^2 y}{dx^2} = M_{BC} = \frac{W}{2} x$$

$$M_{CA} = \frac{W x}{2} - W(x - \frac{L}{2})$$

$$EI \frac{d^2 y}{dx^2} = m = \frac{w}{2} x$$

$$EI \frac{dy}{dx} = \frac{wx^2}{4} + C_1$$

$$EI y = \frac{wx^3}{12} + C_1 x + C_2$$

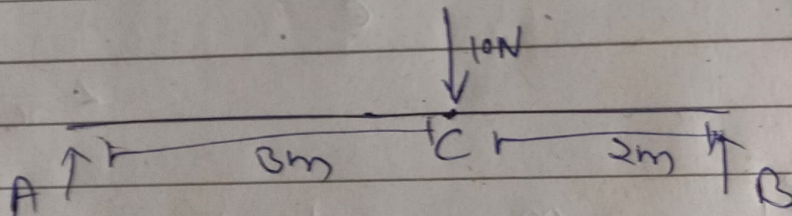
Boundary eq

$$x=0, \quad y=0$$

$$x=L/2, \quad y=0$$

$$x=L/2 \quad y=0 \quad \text{slope} \left(\frac{dy}{dx} \right) = 0$$

Q.



$$R_B \times 5 - 10 \times 3 = 0$$

$$5R_B - 30 = 0$$

$$R_B = 6 \text{ N}$$

$$R_A + R_B = 10 \text{ N}$$

$$R_A = 4 \text{ N}$$

$$EI \frac{d^2 y}{dx^2} = m = 6x$$

$$EI \frac{dy}{dx} = \frac{6x^2}{2} + C_1$$

$$EI y = \frac{6x^3}{6} + \frac{6x}{1} + C_2$$

MACALAY'S method.

Q.3

A open belt drive is required to transmit 10 kW of Power from a motor running at 600 rpm Diameter of the driving pulley is 250 mm The speed of the driven pulley is 220 rpm. The belt is 12 mm thick and has a mass density of 0.001 g/mm³ Safe Stress in the belt is not to

$$D_1 = 250 \text{ mm}$$

$$N_1 = 600 \text{ rpm}$$

$$T = T_1 - T_2$$

Calculation of

Q.4

2.5 kW of power is transmitted by an open belt drive. The linear velocity of the belt is 25 m/sec. The angle of lap on the smaller pulley is 165° . The coefficient of friction is 0.3. Determine the effect on power.